

Determinants of Inverses, Products, and Transposes

Theorem

An $n \times n$ matrix A is invertible if and only if $\det A \neq 0$. In this case,

Statement 1

Statement 2

$$\det(A^{-1}) = \frac{1}{\det A}.$$

The fact that $\det A^{-1} = \frac{1}{\det A}$ will be shown after discussing $\det(AB) = \det A \det B$.

Statement ① $\implies$ Statement ②

If $A_{n \times n}$ is invertible, then we know that $[A|I] \xrightarrow{\text{Elementary operations}} [I|A^{-1}]$. We also know that elementary operations are reversible. Hence A can be obtained from I by performing a sequence of elementary operations. We conclude that $\det A = \alpha \det I = \alpha$, where $\alpha \in \mathbb{R}$ and $\alpha \neq 0$.

Statement ② $\implies$ Statement ①: We show that if S_1 is NOT true, then S_2 is NOT true.

If A is NOT invertible, then A must have less than n pivot positions. This means the RREF will have at least one zero on its main diagonal and since the RREF is an upper triangular matrix, its determinant is the multiplication of the entries on the main diagonal and hence $\det A = 0$.

Statement #1 :

(T/F)

If A and B are two square matrices and A is not invertible then AB is not invertible. True.

We show that if AB is invertible, then A is invertible. Note that if AB is invertible, then there exists a matrix C such that $(AB)C = I$, which means $A(BC) = I$ or $A^{-1} = BC$.

Statement #2 :

(T/F)

For any elementary matrix E and square matrix M , we have:

$$\det(EM) = \det(E) \det(M).$$

True

(I) If E represents exchange of two rows, then $\det E = -1$. Also, $\det(EM) = -\det M$, since the product EM is equivalent to exchange of two rows. Hence, $\det(EM) = \det E \det M$.

(II) If E represents the multiplication of i th row by a constant k , then $\det E = k$ and $\det(EM) = k \det M = \det E \det M$.

(III) If E represents adding k times row i to row j , then $\det E = 1$. and $\det(EM) = \det M = \det E \det M$.

Theorem

Let A and B be $n \times n$ matrices. Then

$$\det(AB) = \det A \times \det B.$$

Case (I): A is not invertible.

If A is invertible, then $\det A \neq 0$. Also, if A is not invertible, then AB is not invertible and $\det(AB) = 0$. Hence, in this case, $\det(AB) = \det A \det B$.

Case (II): A is invertible. Hence, A can be written as a product of EMs.

$$A = E_1 E_2 \dots E_p$$

and

$$\det(AB) = \det(E_1 E_2 \dots E_p B).$$

Next, we can use the statement #2 proved on the previous page to write:

$$\begin{aligned} \det(AB) &= \det E_1 \det(E_2 \dots E_p B) \\ &= \det E_1 \det E_2 \det(E_3 \dots E_p B) \\ &\vdots \\ &= \det E_1 \det E_2 \det E_3 \det E_4 \dots \det E_{p-1} \det E_p \det B \\ &= \det(E_1 E_2) \det E_3 \det E_4 \dots \det E_{p-1} \det E_p \det B \\ &= \det(E_1 E_2 E_3) \det E_4 \dots \det E_{p-1} \det E_p \det B \\ &\vdots \\ &= \det(E_1 E_2 E_3 \dots E_{p-1} E_p) \det B \\ &= \det A \det B. \end{aligned}$$

Corollary: If A is invertible, then $\det A^{-1} = \frac{1}{\det A}$.

Since $AA^{-1} = I$, then $\det(AA^{-1}) = \det A \det A^{-1} = \det I = 1$, hence:

$$\det A^{-1} = \frac{1}{\det A}.$$

Problem

Suppose A , B and C are 4×4 matrices with $\det A = -1$, $\det B = 2$, and $\det C = 1$. Find $\det(2A^2(B^{-1})(C^T)^3B(A^{-1}))$.

$$\begin{aligned}\det(2A^2(B^{-1})(C^T)^3B(A^{-1})) &= (2^4) \det(A^2) \det(B^{-1}) \det((C^T)^3) \det(B) \det(A^{-1}) \\ &= (2^4) (\det A)^2 \frac{1}{\det(B)} (\det(C))^3 \det(B) \frac{1}{\det(A)} \\ &= (2^4) (-1)(1)^3 = -16\end{aligned}$$

Problem

Let $A = \begin{bmatrix} 5 & 1 & 2 \\ 1 & 3 & 2 \\ 2 & 6 & 0 \end{bmatrix}$. Without using the cofactor expansion, find $\det(A)$.

$$A = \begin{bmatrix} 5 & 1 & 2 \\ 1 & 3 & 2 \\ 2 & 6 & 0 \end{bmatrix} \longrightarrow B_1 = \begin{bmatrix} 1 & 3 & 2 \\ 5 & 1 & 2 \\ 2 & 6 & 0 \end{bmatrix} \longrightarrow B_2 = \begin{bmatrix} 1 & 3 & 2 \\ 0 & -14 & -8 \\ 0 & 0 & -4 \end{bmatrix}$$

$$\det A = -\det B_1 = -\det B_2 = -(1)(-14)(-4) = -56$$